



INDIAN INSTITUTE OF INFORMATION TECHNOLOGY RAICHUR  
**MATHEMATICS CLUB**

PROBLEM NUMBER: WP2504

DATE: NOVEMBER 30, 2025

DUE DATE: DECEMBER 15, 2025

## WEEKLY PROBLEM

**EXERCISE.** Let  $f : [0, 2] \rightarrow \mathbb{R}$  be a continuous function satisfying the following conditions:

$$f(0) = 0$$

$$f(1) = 1$$

$$f(2) = 0$$

$$f(x) \geq 0 \quad \text{for all } x \in [0, 2]$$

Assume that the derivative  $f'(x)$  exists except at finitely many points. Let the function  $I(f)$  be defined as:

$$I(f) = \int_0^2 \sqrt{1 + (f'(x))^2} \, dx$$

- (A) Find the minimum possible value of  $I(f)$  over all such functions  $f$ .
- (B) Determine all functions  $f$  for which this minimum is achieved.
- (C) Give a geometric explanation for why your answer is the minimum.



(Scan this to submit solution!)